

CHALLENGE 02

Housing Production, Rents & Household Flow Observatory

Build an interactive regional observatory that shows how housing production, rents, and household movement change over time.

HOW TO USE THESE BRIEFS

These briefs consolidate the original challenge set and the additional ideas developed in review. Each is broad enough to support multiple approaches while still naming a concrete user, decision, and demo.

Teams should build a working prototype, use real regional data where possible, explain assumptions and uncertainty, and demonstrate the product with at least one

Pittsburgh or Allegheny County use case.

CORE PROBLEM

Regional housing decisions are made with fragmented and lagged information. Policymakers cannot easily see what homes were proposed, permitted, completed, rented, or lost; residents lack a shared record of rents and landlord performance; and planners struggle to understand where households of different incomes and sizes are moving.

BUILD CHALLENGE

Build an interactive regional observatory that combines administrative, market, and responsibly crowdsourced data to show how the housing system changes over time. Users should be able to move from a regional pattern to a tract, parcel, building, or project where the underlying data permit. The tool should make missingness, reporting bias, and privacy protections visible.

SUCCESS LOOKS LIKE

A policymaker can answer how many homes were built, where, for whom, and at what apparent rent or price; a resident can contribute or retrieve useful information without exposing personal data; and every visualization communicates data quality.

Owner / input

Planning/data leads with university research support.

Primary users

- County and municipal policymakers tracking housing production
- Researchers and advocates evaluating whether policy is working
- Residents comparing rents, locations, and landlord experiences
- Developers and nonprofits identifying unmet demand

Prototype possibilities

- Housing production dashboard: proposed, permitted, completed, converted, and demolished units
- Rents aggregator with privacy protection, verification, and landlord or property reviews
- Household flow map by income, household size, tenure, origin, and destination
- Housing demand map using prices, rents, income, population, vacancies, and accessibility
- Policy timeline that lets users compare production before and after zoning, tax, or permitting changes

Useful data

- ACS
- HUD CHAS
- County sales and assessment records
- Occupancy certificates
- Documented market indicators

READY TO BUILD?

Use this brief to make regional housing change easier to inspect and explain.

The strongest submissions will help users move from a regional trend to the underlying data while clearly communicating uncertainty and privacy limits.

REGISTER FOR THE HACKATHON

AI Horizons Pittsburgh
AI for Housing Hackathon

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